

Analaysis of Graphene Hall Bar

by Boris

May 13, 2026

1 Landau Fan

1.1 Plotting R_{xx} and R_{xy}

Here is a plot of R_{xx} and R_{xy} :

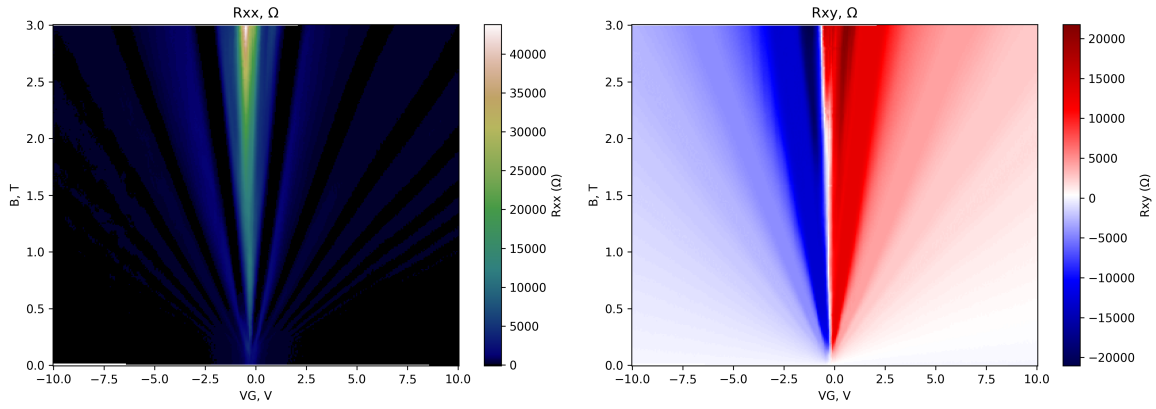


Figure 1: R_{xx} and R_{xy}

Dark lines correspond to Landau levels.

$$n = \nu \frac{eB}{h}; \quad n \propto V_G$$

so the steepest line corresponds to the highest ν . $\nu \in [\pm 2, \pm 6, \pm 10 \dots]$ because of spin-valley degeneracy of monolayer graphene.

1.2 Correlation between n and V_G

To find the correlation $V_G(n)$ i consider two neighboring LL lines with indices ν and $\nu+4$. $V(n) = \alpha B$, with $\alpha_\nu = 2.1 \text{ V/T}$, $\alpha_{\nu+4} = 3.53 \text{ V/T}$ ($\Delta\alpha \approx 1.35 \text{ V/T}$). Plugging in $B(n)$ from the first part:

$$\begin{cases} V_G(n) = \alpha_\nu \frac{nh}{\nu e} \\ V_G(n) = \alpha_{\nu+4} \frac{nh}{(\nu+4)e} \end{cases} \implies V_G(n) = \frac{\Delta\alpha}{4} \frac{nh}{e}$$

$$C = \frac{Q}{V} = \frac{nAe}{V} \approx \frac{4Ae^2}{\Delta\alpha h}$$

$$C = \varepsilon_r \varepsilon_0 \frac{A}{d}; \quad \text{For SiO}_2 \varepsilon_r \approx 3.9$$

$$d = \frac{\varepsilon_r \varepsilon_0 \cdot \Delta\alpha h}{4e^2} \approx 0.3 \mu\text{m}$$

So it probably was a 285 nm waver.

1.3 $R_{xx}(\nu, B)$

Based on the stuff above we can easily compute ν for from B and V_G .

$$\nu = \frac{hn}{eB} = \frac{4V_G}{\Delta\alpha B}$$

One thing is that since $\nu = \frac{hn}{eB}$ it diverges at $B = 0$. I only analyze lines with $|\nu| \leq 10$. I also implement gaussian smoothing on top to smooth things up a bit.

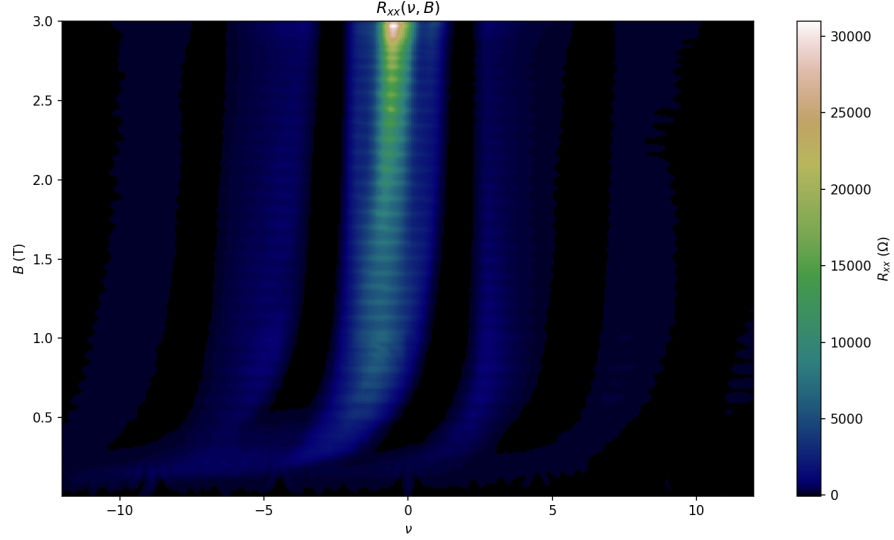


Figure 2: $R_{xx}(\nu, B)$

Vertical dark lines (well, almost vertical) represent landau levels at allowed fillings ν . $\nu \in [\pm 2, \pm 6, \pm 10]$. Like it should.

The lines begin to blur around $B_q \approx 0.4T$.

$$\mu_q B_q = 1 \Rightarrow \boxed{\mu_q \approx 2.5 \text{ m}^2/V \cdot \text{s}}$$

1.4 Symmetry breaking

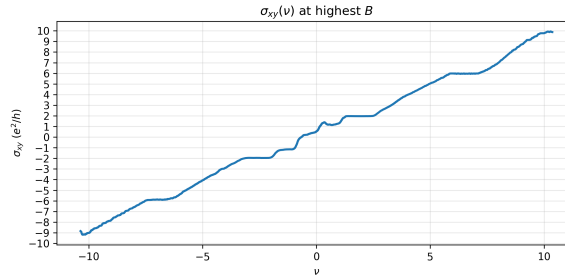


Figure 3: $\sigma_{xy}(\nu)$ at highest $B = 3 \text{ T}$.

Besides plateaus at ν . $\nu = \pm 2, \pm 6, \pm 10$ there are deviations from linear trend at $\nu = \pm 1$ and kinda small bumps around $\nu = \pm 3$ indicating symmetry breaking. The symmetry breaks because of Zeeman splitting, killing spin-degeneracy. $\nu = \pm 3$ bumps are likely results of local time-reversal symmetry breaking. Maybe, due to inter-valley scattering.

2 Shubnikov-de Haas oscillations

2.1 Carrier density

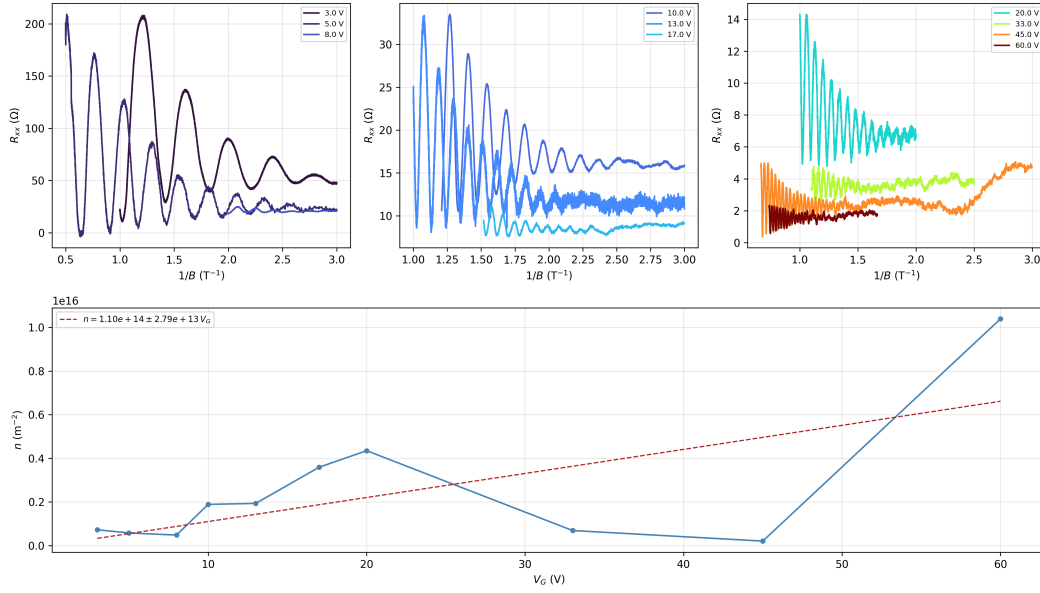


Figure 4: R_{xx} for different V_G and $n(V_G)$.

We can get $n(V_G)$ from SdH oscillations via:

$$n = (2e/h)F$$

Where F is a dominant frequency. To clean the FFT a bit I've filtered the data so that we only work with $1/B < 3 T^{-1}$. This led to more-or-less linear $n(V_G)$ with a coefficient $\beta \approx (1.1 \pm 0.3) \cdot 10^{14} m^{-2} V^{-1}$. Based on the previous part, $\beta_{prev} \approx 7 \cdot 10^{14} m^{-2} V^{-1}$. I'm more inclined to trust the latter. Also, I think that probably the data points corresponding to small V_G are a bit less reliable and they may pull β down. I'll talk about it at the end.

2.2 Quantum lifetime

SdH oscillations decay with $1/\tau_q$ rate:

$$R_{xx} \propto \exp\left(-\frac{\pi m^*}{e\tau_q} \cdot \frac{1}{B}\right)$$

For each V_G i find τ_q as a slope of $R_{xx}(V_G)$ in log/norm coordinates. Averaging over all V_G :

$$\tau_q = 0.12 \pm 0.05 \text{ ps}$$

2.3 Onset oscillations

I can estimate onset B based on $1/B$ when SdH oscillations completely die out. It is different for the different V_G but all in all $1/B_{onset} \in [1.3, 3] T^{-1}$. I eye-balled it to be $2 T^{-1}$, which gives us $B_{onset} \approx 0.5 T$. Analogically to 1.3:

$$\mu_q \approx 2 m^2/V \cdot s$$

I would have actually extracted B_{onset} for every V_G and computed an accuracy, but this will be a computation, and the task was to μ it. Plus it is already getting late.

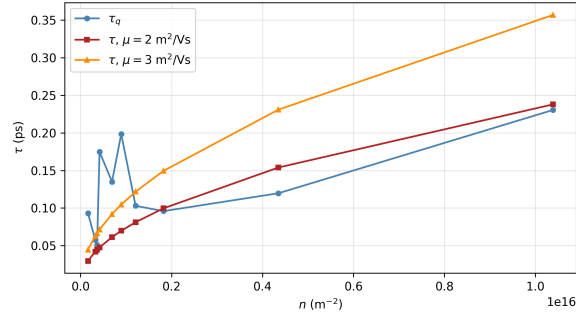


Figure 5: $\tau_q(n)$ and $\tau(n)$ for two different μ

We get τ from Drude model:

$$\mu = \frac{e\tau}{m^*}$$

It ultimately stands for time that it takes to change momentum significantly.

τ_q stands for time between two scattering events.

Ultimately $\tau > \tau_q$. In our case there are two factors i can think of that cause things to look otherwise:

1) my sophisticated eye-balling did eventually let me down (impossible). I've plotted $\tau(n)$ for $\mu = 3 \text{ m}^2/\text{V} \cdot \text{s}$, and it looks a bit better.

2) Estimation of τ_q for small V_G is not a good practice. There are too few oscillations to actually estimate the decay properly, and to extract the dominating frequency.

Anyway, I like the way this plot looks.

This was a fun homework, although it took me way longer than I expected.